

9.1
31 $R_1 = \{(a, b) \mid \text{student } a \text{ is required to read book } b\}$

$R_2 = \{(a, b) \mid \text{student } a \text{ has read book } b\}$

a) $R_1 \cup R_2 = \{(a, b) \mid a \text{ is required to read } b \text{ or } a \text{ has read } b\}$

b) $R_1 \cap R_2 = \{(a, b) \mid a \text{ is required to read } b \text{ and } a \text{ has read } b\}$

c) $R_1 \oplus R_2 = \{(a, b) \mid a \text{ is required to read } b \text{ or } a \text{ has read } b, \text{ but not both}\}$

d) $R_1 - R_2 = \{(a, b) \mid a \text{ is required to read } b \text{ but } a \text{ has not read } b\}$

e) $R_2 - R_1 = \{(a, b) \mid a \text{ has read } b \text{ but } a \text{ is not required to read } b\}$

9.1
33 $R = \{(a, b) \mid a \text{ is a parent of } b\}$

$S = \{(a, b) \mid a \text{ and } b \text{ are siblings}\}$

$S \circ R = \{(a, c) \mid a \text{ is the parent of a sibling of } c\}$

$R \circ S = \{(a, c) \mid a \text{ is the sibling of a parent of } c\}$

9.1
34h $R_2 = \{(a, b) \in \mathbb{R}^2 \mid a \geq b\}$

$R_4 = \{(a, b) \in \mathbb{R}^2 \mid a \leq b\}$

$R_2 \oplus R_4 = \{(a, b) \mid a \geq b \text{ or } a \leq b \text{ but not both}\}$

$= \{(a, b) \mid a > b \text{ or } a < b\}$

$= \{(a, b) \mid a \neq b\} = \boxed{R_6}$

9.1
36a

$$R_1 = \{(a, b) \in \mathbb{R}^2 \mid a > b\}$$

$$R_1 \circ R_1 = \{(a, c) \mid a > b \text{ and } b > c\} = \{(a, c) \mid a > c\} = \boxed{R_1}$$

9.1
37

a) $R_2 \circ R_1 = \{(a, c) \mid a > b \text{ and } b \geq c\} = \{(a, c) \mid a > c\} = \boxed{R_1}$

b) $R_2 \circ R_2 = \{(a, c) \mid a \geq b \text{ and } b \geq c\} = \{(a, c) \mid a \geq c\} = \boxed{R_2}$

c) $R_3 \circ R_5 = \{(a, c) \mid a = b \text{ and } b < c\} = \{(a, c) \mid a < c\} = \boxed{R_3}$

d) $R_4 \circ R_1 = \{(a, c) \mid a > b \text{ and } b \leq c\} = \boxed{\mathbb{R}^2}$ (can choose arbitrarily small b)

e) $R_5 \circ R_3 = \{(a, c) \mid a < b \text{ and } b = c\} = \{(a, c) \mid a < c\} = \boxed{R_3}$

f) $R_3 \circ R_6 = \{(a, c) \mid a \neq b \text{ and } b < c\} = \boxed{\mathbb{R}^2}$ (can choose arbitrarily small b)

g) $R_4 \circ R_6 = \{(a, c) \mid a \neq b \text{ and } b \leq c\} = \boxed{\mathbb{R}^2}$ (can choose arbitrarily small b)

h) $R_6 \circ R_6 = \{(a, c) \mid a \neq b \text{ and } b \neq c\} = \boxed{\mathbb{R}^2}$ (can always pick b to be different from a and c)

9.1
39

$$S_1^2 = \{(a, c) \mid a > b \text{ and } b > c\} = \boxed{\{(a, c) \mid a > c + 1\}}$$

$$S_2^2 = \{(a, c) \mid a \geq b \text{ and } b \geq c\} = \{(a, c) \mid a \geq c\} = \boxed{S_2}$$

$$S_3^2 = \{(a, c) \mid a < b \text{ and } b < c\} = \boxed{\{(a, c) \mid a < c - 1\}}$$

$$S_4^2 = \{(a, c) \mid a \leq b \text{ and } b \leq c\} = \{(a, c) \mid a \leq c\} = \boxed{S_4}$$

$$S_5^2 = \{(a, b) \mid a = b \text{ and } a = c\} = \{(a, c) \mid a = c\} = \boxed{S_5}$$

$$S_6 = \{(a, b) \mid a \neq b \text{ and } b \neq c\} = \boxed{\mathbb{Z}^2}$$
 (can always pick b to be different from a and c)

9.1
41 $R = \{ (a, b) \mid a \text{ was the thesis advisor of } b \}$

$R^2 = \{ (a, b) \mid a \text{ was the thesis advisor of a thesis advisor of } b \}$

$R^* = \{ (a, b) \mid a \text{ was the thesis advisor to some sequence of thesis advisors of } b \}$

9.1
43 $R_1 = \{ (a, b) \mid a \equiv b \pmod{3} \}$

$R_2 = \{ (a, b) \mid a \equiv b \pmod{4} \}$

a) $R_1 \cup R_2 = \{ (a, b) \mid a \equiv b \pmod{4} \text{ or } a \equiv b \pmod{3} \}$

b) $R_1 \cap R_2 = \{ (a, b) \mid a \equiv b \pmod{4} \text{ and } a \equiv b \pmod{3} \} = \{ (a, b) \mid a \equiv b \pmod{12} \}$

c) $R_1 - R_2 = \{ (a, b) \mid a \equiv b \pmod{3} \text{ but } a \not\equiv b \pmod{4} \}$

d) $R_2 - R_1 = \{ (a, b) \mid a \equiv b \pmod{4} \text{ but } a \not\equiv b \pmod{3} \}$

e) $R_1 \oplus R_2 = \{ (a, b) \mid a \equiv b \pmod{3} \text{ or } a \equiv b \pmod{4} \text{ but not both} \}$

9.2
1 $\{ (a, b, c) \mid a, b, c \in \mathbb{Z} \text{ with } 0 < a < b < c < 5 \}$

$(1, 2, 3), (1, 2, 4), (1, 3, 4), (2, 3, 4)$

9.2
2 $\{ (a, b, c, d) \mid a, b, c, d \in \mathbb{Z}^+ \text{ and } abcd = 6 \}$

$(1, 1, 1, 6)$	$(1, 1, 2, 3)$	$(1, 1, 3, 2)$	$(1, 2, 1, 3)$	$(1, 3, 1, 2)$
$(1, 1, 6, 1)$	$(1, 2, 3, 1)$	$(1, 3, 2, 1)$	$(2, 1, 3, 1)$	$(3, 1, 2, 1)$
$(1, 6, 1, 1)$	$(2, 3, 1, 1)$	$(3, 2, 1, 1)$	$(2, 1, 1, 3)$	$(3, 1, 1, 2)$
$(6, 1, 1, 1)$				

9.2
3 (Nadir, 122, 34, -Detroit, 08:10)

rest of the lines of the table

9.2
5 (Airline, Flight Number) since Flight Number is different for each row

9.2
9 a) Social Security number since it is unique for each person
b) No two people have the same name and street address
c) No two people with the same name living in the same place

9.2
10 Each row with A100 as their room (first 4 rows).

9.2
15 $P_{3,5,6}$ (omit 1, 2, 4)

9.2
28b 23, 31, 33

9.2
31 no, none of the fields can determine the other fields

9.3
2c
$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

- 9.3
3
- a) $(1,1), (1,3), (2,2), (3,1), (3,3)$
- b) $(1,2), (2,2), (3,2)$
- c) $(1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (3,3)$

9.3
5

A relation is irreflexive if and only if its matrix has all 0s on the diagonal.

9.3 7		reflexive	irreflexive	symmetric	antisymmetric	transitive
a)		✓	X	✓	X	✓
b)		X	X	X	✓	✓
c)		X	X	✓	X	X
9.3 8						
a)		X	X	✓	X	X
b)		✓	X	X	✓	X
c)		X	✓	✓	X	X

9.3
9 a) $\{(a,b) \mid a > b\}$

$$1 + 2 + \dots + 99 = \frac{99(100)}{2} = \boxed{4950}$$

b) $\{(a,b) \mid a \neq b\}$

$$99 \cdot 100 = \boxed{9900}$$

c) $\{(a,b) \mid a = b+1\}$

$$\boxed{99}$$

d) $\{(a,b) \mid a = 1\}$

$$\boxed{100}$$

e) $\{(a,b) \mid ab = 1\}$

$$\boxed{1}$$

9.3
10a $\{(a,b) \mid a \leq b\}$ $1 + 2 + 3 + \dots + 1000 = \frac{1000(1001)}{2} = \boxed{500500}$

9.3
12 Take the transpose of the matrix

9.3
13

a) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

c) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

9.3
14

a) $R_1 \cup R_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

b) $R_1 \cap R_2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

c) $R_2 \circ R_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

d) $R_1 \circ R_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

e) $R_1 \oplus R_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

$$\textcircled{9.3} \quad \text{a)} \quad \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\text{b)} \quad \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\text{c)} \quad \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\textcircled{9.4} \quad \{(0,1), (1,1), (1,2), (2,0), (2,2), (3,0)\}$$

$$\text{a)} \quad \{(0,0), (0,1), (1,1), (1,2), (2,0), (2,2), (3,0), (3,3)\}$$

$$\text{b)} \quad \{(0,1), (0,2), (0,3), (1,0), (1,1), (1,2), (2,0), (2,1), (2,2), (3,0)\}$$

$$\textcircled{9.41} \quad R = \{(a,b) \mid a \neq b\}$$

add all pairs $(x,x) \Rightarrow \mathbb{Z} \times \mathbb{Z}$ (all possible pairs)